

EE2080: Problem Set 3: Discrete-Time Markov Chains

Sample Applications

1. PageRank: Ranking Web Pages

A website consists of four pages A, B, C, D . Page A links to B and C ; B links to C ; C links to A ; and D links to A and C . A user on a page chooses one of its outgoing links uniformly at random.

- Model the sequence of webpages visited by the user as a time-homogeneous DTMC and write its transition matrix.
- If the user starts at A , what is the probability that the user is on page C after three clicks?
- Which pages, once left, can never be visited again? Which groups of pages can the user continue moving among indefinitely?
- Suppose that a page is ranked according to the fraction of time a random user spends on it in the long run. Determine the ranking of the four pages.

A PageRank-type model modifies the browsing behaviour as follows. At each step, with probability 0.85 the user follows a randomly chosen outgoing link, while with probability 0.15 the user jumps to one of the four pages uniformly at random.

- Determine the transition probabilities under the new browsing rule.
- Explain why, under the new rule, it is possible to eventually get from any page to every other page.
- Determine the long-run fraction of time spent on each page and hence rank the pages.
- Explain the role played by the random jump in determining the ranking.

2. Search Engine Web Crawler

A web crawler moves among four webpages according to the transition matrix

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Page 3 contains a document that the crawler is trying to locate.

- If the crawler starts at page 0, what is the probability that it eventually finds the document?
- Starting from each of pages 0, 1, 2, calculate the expected number of page transitions required to find the document.
- Suppose the crawler has just reached page 2 for the first time. What is the probability that it finds the document within its next two transitions?
- Does knowing the entire route by which the crawler first reached page 2 change your answer to part (c)? Explain.
- Starting from page 0, let q_n be the probability that the crawler finds the document on page 3 *for the first time* after exactly n page transitions. Find q_1, q_2, q_3 , and obtain a recurrence that can be used to determine q_n for $n \geq 4$.
- Explain the difference between q_n and the probability that the crawler is on page 3 after n transitions. How are the two quantities related in this particular model?

3. Server Reliability

A server is classified at the end of each day as operational (O), degraded (D), or failed (F). Its daily evolution is described by

$$P = \begin{pmatrix} 0.9 & 0.1 & 0 \\ 0.4 & 0.4 & 0.2 \\ 0.8 & 0 & 0.2 \end{pmatrix},$$

where the states are ordered as O, D, F .

- (a) A new server is operational today. What is the probability that it is failed two days from now?
- (b) Starting with an operational server, calculate the expected number of days before its first failure.
- (c) Starting with an operational server, let q_n denote the probability that the server fails for the first time on day n . Find q_1, q_2 , and q_3 .
- (d) Obtain an expression or recurrence for $q_n, n \geq 1$. Hence verify that the expected time to first failure obtained from this distribution is consistent with your answer to part (b).
- (e) Over several years of operation, what fraction of days would you expect the server to be operational, degraded, and failed?
- (f) A failed server costs Rs. 50,000 per day and a degraded server costs Rs. 10,000 per day in lost productivity. Estimate the long-run average loss per day.
- (g) An improved repair system changes the probability of moving from F to O from 0.8 to 0.95, with the probability of remaining in F changing accordingly. Determine how this affects the long-run average loss.

4. Video Recommendation System

A recommendation system classifies a user's engagement after each recommended video as low (L), medium (M), or high (H). Under a particular recommendation policy,

$$P = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.3 & 0.6 \end{pmatrix},$$

where the states are ordered as L, M, H .

- (a) A new user begins with low engagement. What is the probability that the user's engagement is high after three recommendations?
- (b) Starting with low engagement, how many recommendations are required on average before the user becomes highly engaged for the first time?
- (c) A user has just become highly engaged for the first time. What is the probability that the user's engagement is still high two recommendations later?
- (d) Would the answer to part (c) change if you were also told that the user took exactly seven recommendations to become highly engaged? Explain.
- (e) Starting with low engagement, let q_n denote the probability that the user becomes highly engaged for the first time immediately after the n th recommendation. Find q_1, q_2 , and q_3 .
- (f) Give a method for computing q_n for arbitrary n . Using your answer, find the probability that a user starting with low engagement becomes highly engaged for the first time within the first three recommendations.
- (g) Compare the probability in part (f) with the probability that the user's engagement is high after exactly three recommendations. Explain why these quantities need not be equal.
- (h) For a user who uses the platform for a very long time, determine the expected fraction of recommendations after which the user's engagement is high.

5. Communication Network

Four routers A, B, C, D are connected in a ring, $A - B - C - D - A$. A diagnostic packet arriving at a router is forwarded to either neighbouring router with equal probability.

- (a) Model the location of the packet as a time-homogeneous DTMC and write its transition matrix.

- (b) Starting from A , where can the packet be after 1001 transitions? Can it be at A ? Explain.
- (c) Starting at A , determine the fraction of time that the packet spends at each router in the long run.
- (d) An engineer observes that although the fractions in part (c) are well defined, the probability that the packet is at A after n transitions does not approach the corresponding fraction as $n \rightarrow \infty$. Explain why.

The routing protocol is changed so that at every step the packet remains at its current router with probability 0.1 and is forwarded to either neighbour with probability 0.45 each.

- (e) Does the long-run fraction of time spent at each router change?
- (f) What important difference does this modification make to the long-term movement of the packet?

6. Wireless Packet Transmission

A wireless transmitter is allowed at most three attempts to deliver a packet. Each transmission succeeds independently with probability p . If all three attempts fail, the packet is dropped.

- (a) Construct a time-homogeneous DTMC that describes the status of the packet after each transmission attempt. Clearly specify the states and draw the transition diagram.
- (b) What is the probability that a packet is successfully delivered?
- (c) What is the expected number of transmissions used per packet?
- (d) Among packets that are successfully delivered, what is the probability that more than one transmission was required?
- (e) Suppose each transmission consumes E joules of energy. Determine the expected energy consumed per packet.
- (f) If successful delivery earns a reward of R and every transmission costs c , determine the expected net reward per packet.

7. Customer Switching Between Services

Customers choose among three mobile service providers A, B, C . Their monthly switching behaviour is described by

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

Assume that customers behave independently according to this model.

- (a) Of one million customers currently subscribed to A , how many would you expect to be subscribed to C two months from now?
- (b) If this switching behaviour continues for many years, estimate the market share of each provider.
- (c) Provider C introduces a loyalty programme that increases the probability that a current C subscriber remains with C from 0.5 to 0.8. The probabilities of switching from C to A and B are reduced proportionally. Estimate the new long-run market shares.
- (d) Which provider benefits or loses the most from this change?

8. Online Game

A player's rating level is one of $0, 1, \dots, 10$. The player starts at level 4. After each game, the player's level increases by one with probability $1/2$ and decreases by one with probability $1/2$. Reaching level 10 earns the player the highest rank, while reaching level 0 eliminates the player from the competition. The process ends upon reaching either 0 or 10.

- (a) What is the probability that the player earns the highest rank?
- (b) How many games does the competition last on average for this player?
- (c) What is the probability that the player reaches level 7 at least once before being eliminated?
- (d) Suppose you are informed that the player has just reached level 7 for the first time. What is the probability that the player subsequently reaches level 10 before being eliminated?
- (e) Hence determine the probability that a player starting at level 4 reaches level 7 and subsequently achieves the highest rank.

9. Server Monitoring and Decision Times

A server is inspected once every hour and its state X_n is recorded as operational (O), degraded (D), or failed (F). The sequence $\{X_n\}_{n \geq 0}$ is modelled as a time-homogeneous DTMC.

An administrator considers three alarm rules: T_1 is the first hour at which the server is found to have failed; T_2 is the last hour during the first 24 hours at which the server is found to have failed; and T_3 is the first hour at which the server is operational and remains operational for the following three hours.

- (a) For each rule, determine whether the administrator can know, using only the observations available up to the current hour, that the specified time has occurred.
- (b) Hence determine which of T_1, T_2, T_3 are stopping times with respect to the observations $\{X_0, X_1, \dots\}$.
- (c) For each time that is not a stopping time, explain what future information is required in order to identify it.

10. Movement Between Network Subsystems

A computer network contains six routers A, B, C, D, E, F . From A , a packet moves to B with probability 1. From B , it moves to A or C , each with probability $1/2$. From C , it moves to A with probability $1/2$ and to D with probability $1/2$. From D , it moves to E with probability 1. From E , it moves to D or F , each with probability $1/2$. From F , it moves to D with probability 1.

- (a) Draw the transition graph.
- (b) Determine which routers can be reached from each other in both directions. Hence divide the network into groups of mutually accessible routers.
- (c) Which of these groups can a packet leave without being able to return?
- (d) Suppose it is known that a packet starting at router D returns to D infinitely often with probability 1. What can you conclude about repeated visits to routers E and F ? Justify your answer.
- (e) Suppose it is known that returns to D can occur only after an even number of transitions. What can you conclude about the possible return times to E and F ?
- (f) Classify the routers according to whether a packet starting there is certain to return eventually or has a positive probability of never returning.